

The correlation of glycemic control, apolipoprotein A1 and lipid peroxidation with dyslipidemia, in type 2 diabetes patients is age dependent

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Abstract:

Introduction: Diabetes is a serious, long-term condition with a major impact on the lives and well-being of individuals, families, and societies worldwide. Glycemic control is an essential strategy for the correction of lipid metabolism in order to decrease diabetes complications. This work aimed to compare the correlation of glycemic control with lipid metabolism and lipid peroxidation in type 2 diabetes patients in two age groups of adult type 2 DM patients.

Materials and methods: A cross-sectional study of a sample of patients visiting “El Gemeih Center for diabetes care and surgery”, Dongola locality, Sudan. Glycemic control (HbA1c %), lipid panel, lipid ratios, apolipoprotein A1 (major protein constituent of High density lipoprotein molecules) and lipid peroxidation were measured, patients' data were grouped into two age groups; over 60, and 60 y old or younger.

Results and conclusion: Glycemic control correlates positively with total cholesterol, LDLc, lipid ratios and lipid peroxidation, and negatively with apoA1 and HDLc. In older age HbA1c had no correlation with lipid peroxidation or apolipoproteinA1. Lipid peroxidation levels in older patients correlate with lipid profile not the age, contrary to younger adults. HbA1c in younger adults is more significantly correlated with lipid profile normalization and rate of apoA1 synthesis. ApoA1 correlation with lipid indices was dissociated when HbA1c is within controlled limits in older group. Therefore oxidative stress might need more attention in elderly type 2 diabetes patients, while apoA1 and glycemic control are more decisive in younger adult type 2 diabetes patients.

Abbreviations: TCh=Total cholesterol, TG=Triglycerides, AI = Atherogenic Index (Tg/HDLc), HbA1c = Glycosylated hemoglobin, apoA1=apolipoproteinA1

Key words: type 2 diabetes, apolipoproteinA1, dyslipidemia, lipid peroxidation

مستخلص:

يعد السكري مرضاً مزمنًا ذا خطر، وله آثاره الكبيرة على حياة وجودة عيش الأفراد والمجتمعات عالمياً. وتعتبر عملية ضبط مستويات السكري استراتيجية أساسية من أجل تصحيح أيض الدهون بغرض تخفيف الآثار المستقبلية المترتبة على المرض. تهدف هذه التجربة إلى دراسة أثر ضبط السكري على أيض الدهون وأكسدة الفوقية في مجموعتين عمريتين من مرضى السكري من النوع الثاني.

المواد المستخدمة والتجريب: أجريت دراسة مقطعية على عينة من مرضى السكري النوع الثاني المترددين على "مركز الجميع رعاية وجراحات السكري" بمحلية دنقلا، الولاية الشمالية بالسودان. تم قياس السكر التراكمي كنسبة مئوية، صورة الدهون، النسب الدهنية والبروتين الصافي¹ (البروتين الرئيسي في تكوين جزيء الليبوبروتين عالي الكثافة) والأكسدة الفوقية للدهون، ثم قسمت بيانات

المرضي حسب العمر إلى فئتين، الأولى تبلغ ستين عاما أو أقل والثانية أعمارهم تزيد عن الستين. في الفترة بين يناير ونوفمبر 2022.

النتائج والخلاصة: يتناسب ضبط السكر التراكمي طرديا مع الكولسترول الكلي وكولسترول الليبوبروتين الأقل كثافة ونسب الدهون والأكسدة الفوقية وعكسيا مع كولسترول الليبوبروتين عالي الكثافة ومع البروتين الصافي 1. تحت ظروف ضبط السكر التراكمي تظهر المجموعة العمرية الأصغر ترافقا معنويا مع تحسن صورة الدهون ومعدل تخليق البروتين الصافي 1 بينما افتقد هذا الترافق في المجموعة العمرية الأكبر كما يظهر انعدام الترافق مع الأكسدة الفوقية للدهون في المجموعة الأكبر عمريا لذا فربما يجدر الانتباه إلى الأكسدة الفوقية في العمر الأكبر كمتغير مهم لا يرتبط بضبط السكر التراكمي من مرضي السكري النوع الثاني بينما يبدو أن ضبط السكر التراكمي في الأعمار الأصغر له تأثيره على معدل تخليق البروتين الصافي وتحسن صورة الدهون والأكسدة الفوقية.

Introduction:

Type 2 diabetes, on the other hand, includes hyperglycemia due to insulin resistance and insulin secretion deficiency ^[1]. The actual effect of type 2 diabetes is the outcome of a complex physiologic process including peripheral glucose uptake in adipose and muscle tissue, the hormonal / cytokine secretion activity of adipocytes, impaired metabolism of lipids and lipoproteins and hepatic production of glucose and dysfunctional beta cells of the pancreas ^[2]. Glycemic control is tested by levels of glycosylated hemoglobin HbA1c in type 2 DM patients ^[3]. Lipid profiles of diabetic patients is greatly shifted towards atherogenic limits, mostly have high triglycerides (TG), total cholesterol (TCh), low-density lipoprotein cholesterol (LDLc), otherwise, they have low levels of high-density lipoprotein cholesterol ^[4], which greatly increases their risk for atherosclerosis and cardiovascular diseases (CVD). Dyslipidemia caused by prolonged diabetes and insulin resistance and under conditions of poor glycemic control was found to be the major factor in atherosclerosis progression, independent of hyperglycaemia ^[5,6]. Increased amounts of proinflammatory cytokines and oxidative stress are suggested to mediate those side effects of diabetes ^[7]. According to the American Diabetes Association (ADA), standards of medical care-2020 abridged issuance article 6: Glycemic targets, it is stated that "Glycemic management is primarily assessed with the HbA1c test". The acceptable glycemic control in cases where hypoglycemia or adverse effects of treatment is avoided HbA1c of 7% value is considered appropriate, and either 8% level is considered suitable for older people and patients with severe complications, however in middle aged people 6.5 % is still a more desirable criterion ^[8]. In the study of Sreenivas Reddy et al, 2014, a negative significant correlation was found between age and LDLc, TG, and total cholesterol, although the age was not significantly correlated with HbA1c or fasting blood glucose (FBG), besides HbA1c was an indirect indicator of dyslipidemia ^[9]. Oxidative stress is a status of imbalance between total antioxidants and total reactive oxygen species (ROS) in the body ^[10]. Lipid peroxidation as a direct biochemical result of oxidative stress is known to be increased in diabetic patients as compared to healthy individuals, and it is strongly correlated to diabetic complications, such as cardiovascular disease, kidney disease, and neuropathy, retinopathy and foot ulcers ¹¹. It has been documented that diabetes affects HDL turnover as a molecules and demolishes its antioxidant capacity, mainly through reduced synthesis of its apoprotein parent constituent apolipoprotein A1 ^[12, 13]. One of the theories of type 2 diabetes mellitus is that aging is an independent risk factor for the disease due oxidative stress and due to disorganized mitochondria, besides the rate of synthesis of many defensive enzymes might be slowed down in old ages independent of other risk factors such as body mass index (BMI) ^[14,15]. Recently lipid ratios and oxidative stress had arisen as fields of question as better indices or markers for future cardiovascular incidences than absolute values of lipid panel ^[16].

Aim of the work:

To investigate whether glycemic control correlates with lipid panel, lipid ratios, apoA1, and lipid peroxidation differently in younger and older adults.

Methods:

5 ml venous blood samples were collected from patients. Serum was separated and preserved in liquid nitrogen until further analysis.

Lipid profile determination was conducted on serum, according to manufacturer directions, total cholesterol biosystems kits code 11805, HDLc biosystems kits code 11648, Triglyceride biosystems kits code 11828. LDLc was calculated from equation of Martin-Hopkins equation ^[17]. Lipid peroxidation was determined in serum using a modification of dROMs method by Nassar et al, 2020 ^[18], N,N-dimethyl-p-phenylene diamine hydrochloride was purchased from Avondale laboratories, England. Hydrogen peroxide standards S1 0.4, S2 0.8, S3 1.2, S4 1.6, S5 2, S6 2.0, S7 3.0, S8 4 µg/ml prepared from stock 30% from labogend India. We used ELISA reader Mindray MR-96A at primary wavelength 492 nm and 630 nm as secondary wavelength.

apolipoproteinA1 (apoA1) was determined by antigen antibody reaction measured by ELISA using two kits of Quantikine ELISA according to manufacturer R&D systems a biotechne brand, USA, code 894879. Standard curves were obtained and sample concentration estimated automatically from the reader.

Results and Discussion:

People 60 years or older were considered as senior adults of elderly group while people between 18-59 as young adults, according to the Chinese government analysis of population published in 2019 ^[19], or according to the National Institutes of Health (NIH) guideline, people over the age of 65 are considered older adults, while elderly describes people older than 65 ^[20], therefore we opted to choose the line of 60 to describe older people as a unite group to keep in line with the average life expected age in Sudan which is 66 years according to world bank data in 2023 ^[21]. Patients older than 60 year old counted 36 patients, while patients under 60 years old counted 140. Older patients with uncontrolled HbA1c counted 20 (group 1 unc) and those with controlled HbA1c counted 16 patients (group 1 cont.), the younger age group uncontrolled HbA1c (group 2 unc) were 67 patients, the controlled (group 2 cont.) counted 73 patients.

“Uncontrolled HbA1c”:

LDLc/HDLc, HDLc, and lipid peroxidation were significantly age correlated in (group 1 unc), this gets in agreement with previous work where dyslipidemia increases with age in older subjects irrespective of diabetes, where features of dyslipidemia increase with age ^[22,23]. ApoA1 in this group correlated significantly with TCh, HDLc, LDLc, and atherogenic index (AI), which agrees with documented works as apoB/apoAI ratio with nonHDLc in elderly up to 70 years old ^[24]. In this group Lipid panel and lipid ratios were associated with apolipoproteinA1 rather than HbA1c, the logic of this is the documented association between apoA1 and HDL structure and function, furthermore, this study had revealed that there is association between age and lipid peroxidation under hyperglycemia, which means that the levels of oxidative stress are age dependent in older type 2 diabetes patients, meanwhile lipid peroxidation was found associated significantly with lipid profile and lipid ratios in this group which suggests an added role for oxidative stress in the elderly to their dyslipidemia under uncontrolled hyperglycemia conditions. Additional observation is the absence of correlation between apoA1 and lipid peroxidation, which indicates that decreased levels of apoA1 in this group is correlated mainly with aging, in the light of previous literature that proved correlation between lipid peroxidation and age independently with the gene expression of apoA1 ^[25, 26].

(Group 2 unc), results agrees with studies that documented correlation of HbA1c with dyslipidemia in young adults with type 1 diabetes ^[27], while other studies found that type 2 diabetes patients between 18 to 40 years old have association between glycemic control and lipid profile ^[28,29], however this comparison concerning the quality and quantity of HDL as a molecule between younger and older done in this study and revealed less quantity of HDL assessed as apoA1 decrement, and lipid peroxidation increase was in agreement with few previous studies on the efflux properties and composition of HDL besides deteriorated HDL capacity as cholesterol acceptors, in elderly ^[30], while another study found that kinetics of HDL and apoA1 is not function of age in young adults ^[31].

“Controlled HbA1c”:

In (group 2 cont.) <60 years old, HbA1c was correlated with a number of lipid panel and lipid ratios besides its correlation with the degree of lipid peroxidation, this conforms with previously findings in HbA1c to be strong predictor of lipid panel ^[32], other studies found HbA1c predictive only of Triglycerides, in Saudi Arabia, although, the study was not age adjusted with a mean age of 54 years old ^[33], Lipid peroxidation in this young group was strongly correlated with glycemic control and, TG and atherogenic index besides inverse strong correlation with levels of apoA1, the latter might be the pacemaker of HDL quantity while oxidative stress burden plays a major role in HDL functionality. On the other hand (group 1 cont) older patients, the study on elderly from Zhejiang, China, found Triglycerides only be predictive of glycemic control ^[34]. While in our study HbA1c showed correlation with fasting plasma glucose only, this is common observation with uncontrolled HbA1c which indicates that HbA1c is not reliable as a lipid profile predictor in the old age with chronic hyperglycemia, instead, apA1 and lipid peroxidation were correlated with lipid profile and lipid ratios, due to their cross talking roles, lipid peroxidation affects the level and structure of apoA1 and the latter affects directly the levels and structure of HDL. So our results were consistent with documented previous works that relate changes in lipid peroxidation to age in normal population or among diabetes patients which we suggest as added burden to factors that affect dyslipidemia and apolipoprotein and HDL kinetics ^[35, 36].

Recommendations: more research is needed in Sudan and sub-Saharan countries to formulate a more detailed picture of diabetes effects on lipid metabolism, and more attention should be given in older ages to reducing oxidative stress, as it might be independent player.

Tables:

Table 1: Correlation of HbA1c with lipid panel, lipid peroxidation and apoA1 in (group 1 uncon)

		FBG	TCh	HDLc	TGs	LDLc	LDLc/HDLc	AI	Lipid peroxidation	apoA1	Age
HbA1c	Pearson Correlation	0.72**	-0.09	-0.35	0.16	0.04	0.24**	0.28	0.37	-0.36	.093**
	Sig. (2-tailed)	0.00	0.69	0.13	0.50	0.86	0.03	0.26	0.10	0.11	.696

*Significance level <0.05

** significance level <0.01

Table 2: Correlation of ApoA1 with lipid panel and lipid peroxidation in (group 1unc):

		HbA1c	FBG	TCh	HDLc	TG	LDLc	LDLc/HDLc	AI	Lipid peroxidation
ApoA1	Pearson Correlation	-0.36	-0.27	0.56*	0.65**	-0.14	0.45	-0.33	-.52*	-0.33**
	Sig. (2-	0.11	0.30	0.01	0.00	.57	0.04	0.15	.026	0.14

	tailed)										
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*Significance level <0.05

** significance level <0.01

Table 3: Correlation of HbA1c with lipid panel, lipid peroxidation , apoA1 and age in (group 2 unc)

		TCh	HDLc	TGs	LDLc	LDLc/ HDLc	FBS	AI	ApoA	Lipid peroxida tion	Age
HbA1c	Pearson correlation	0.04	-0.29**	0.37**	0.05	0.21	0.45**	0.44**	-0.16	0.42	0.01**
	Sig. (2- tailed)	0.68	0.00	0.00	0.63	0.04	0.00	0.00	0.13	0.00	0.90

*Significance level <0.05

** significance level <0.01

Table 4: Correlation of apoA1 with lipid panel, age and lipid peroxidation in (group 2 unc)

		HbA1c	TCh	HDLc	TGs	LDLc	LDLHDL	FBS	AI	Lipid peroxid ation	Age
apoA1	Pearson Correlatio n	-.160	.364**	.564**	.221*	.206	-.361**	-.180**	-.272*	-.025**	.067**
	Sig. (2- tailed)	.138	.001	.000	.042	.059	.001	.108	.012	.820	.539

*Significance level <0.05

** significance level <0.01

Table 5: Correlation of HbA1c with lipid panel, apoA1a and lipid peroxidation in (group 1 con)

		TCh	HDLc	TG	LDLc	LDL/HDLc	AI	FBG	Lipid peroxide	ApoA
HbA1c	Pearson Correlation	.231	-.773**	.010	.512	.231	.086	.779**	.244	-.193
	Sig. (2- tailed)	.494	.005	.977	.108	.493	.801	.013	.470	.570

*Significance level <0.05

** significance level <0.01

Table 6: Correlation of apoA1 with lipid panel, HbA1c and lipid peroxidation in (group 1 con)

		HbA1c	TCh	HDLc	TG	LDLc	LDLcHDL	AI	FBG	Lipid peroxidation
apoA1	Pearson Correlation	-.193	.093	.632*	.108	-.335	-.601	-.307	-.163*	-.233
	Sig. (2- tailed)	.570	.785	.037	.752	.313	.050	.358	.676	.490

*Significance level <0.05

** significance level <0.01

Table 7: Correlation of HbA1c with lipid panel, apoA1 a and lipid peroxidation in (group 2 con)

		FBG	Tch	HDLc	TGs	LDLc	LDLc/HDLc	AI	apoA	Lipid peroxid ation
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HbA1c	Pearson Correlation	.38**	.10	-.039	.29	.058**	.14	.37	-.56	.48**
	Sig. (2-tailed)	.002	.411	.75	.017	.64	.25	.002	.000	.000

*Significance level <0.05

** significance level <0.01

Table 8: Correlation of apoA1 with lipid panel and lipid peroxidation in (group 2 con)

		HbA1c	TCHOLEST	HDLc	TGs	LDLc	LDL/HDL	FBG	AI	Lipid peroxidation
apoA1	Pearson correlation	-.560**	-.091	.21	-.25*	-.184	-.35**	-.31*	-.42**	-.88**
	Sig. (2-tailed)	.000	.46	.07	.03	.13	.00	.013	.00	.00

*Significance level <0.05

** significance level <0.01

Table 9: Correlation of lipid peroxidation with lipid panel, HbA1c and apoA1 (group 2 con)

		HbA1c	TCh	HDLc	TGs	LDLc	LDLc/HDLc	FBS	AI	ApoA	Age
Lipid peroxidation	Pearson Correlation	.487**	.062	-.155	.242*	.117	.235	.229	.405**	-.886**	.165
	Sig. (2-tailed)	.000	.620	.215	.048	.346	.058	.071	.001	.000	.181

*Significance level <0.05

** significance level <0.01

Figures:

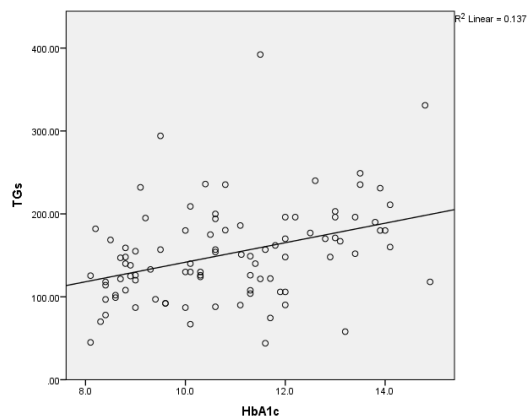


Fig. 1
HbA1c to Tg correlation in group 2 unc

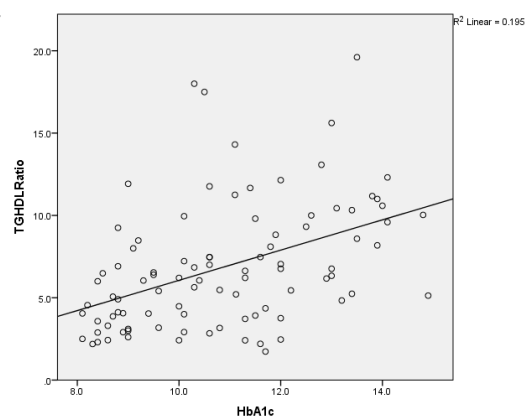


Fig. 2
HbA1c to Tg/HDLc correlation in group 2 unc

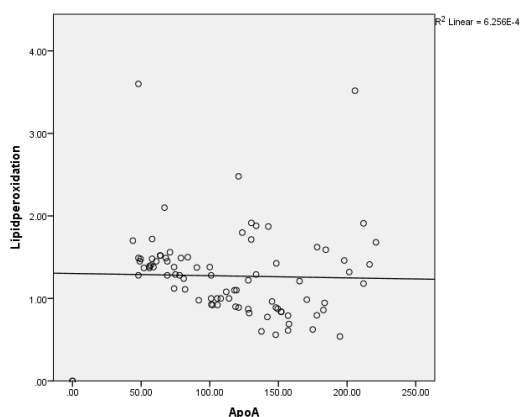


Fig. 3

apoA1 to lipid peroxidation correlation in group 2 unc
unc

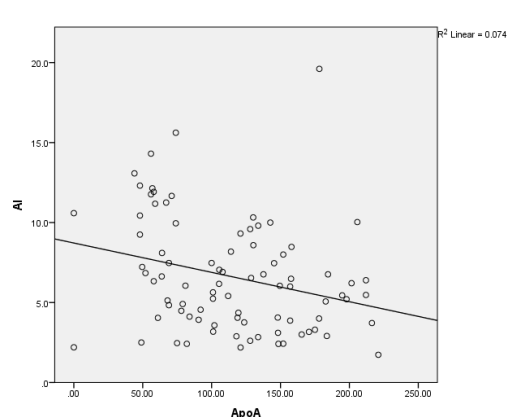


Fig. 4

apoA1 to AI correlation in group 2

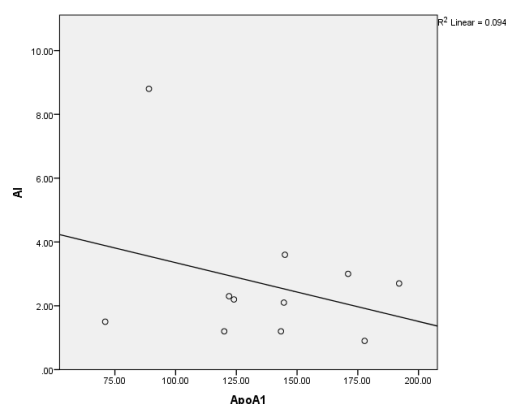


Fig. 5

apoA1 to AI correlation in group 1 cont.

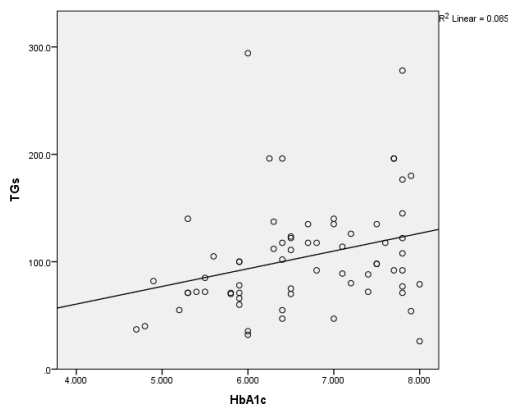


Fig. 6

HbA1c to TG correlation in group2 cont

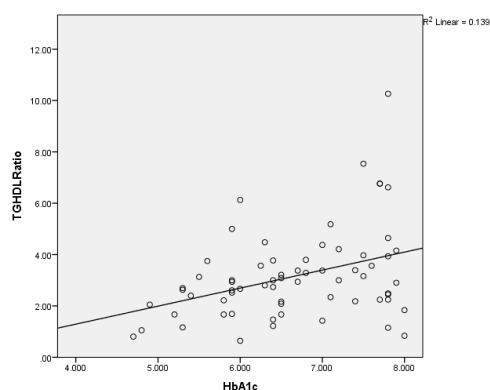


Fig 7

HbA1c to AI correlation in group2 cont

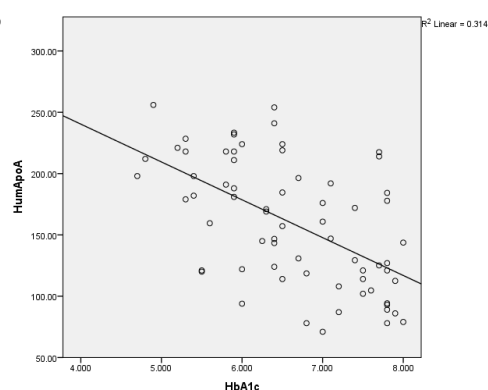


Fig 8

HbA1c to apoA1 correlation in group 2 cont

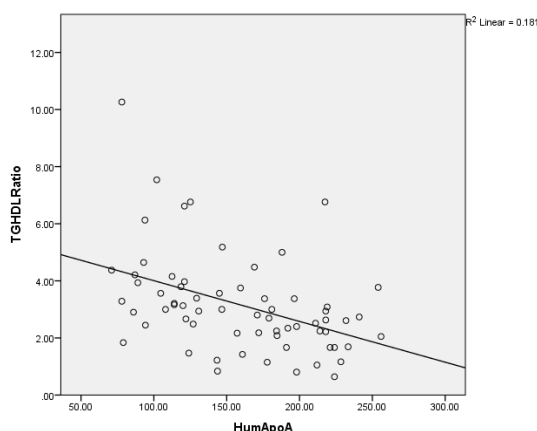


Fig 9

apoA1 to AI correlation in group 2 cont.

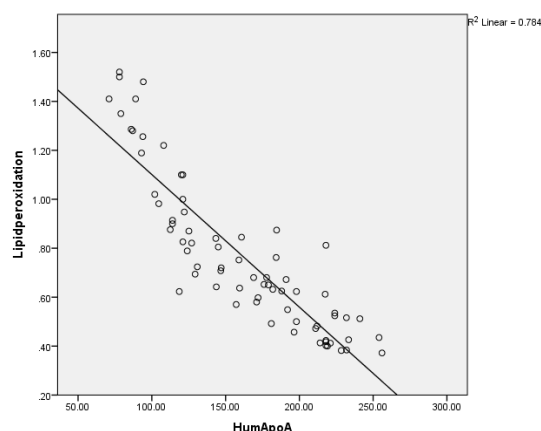


Fig 10

apoA to lipid peroxide correlation group 2 cont.

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